



CSE422: Artificial Intelligence

Group: 09

Section: 07

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Table of Contents

	pages
1. Introduction	3
2. Dataset Description	3
3. Class Distribution & Imbalance	5
4. Exploratory Data Analysis (EDA)	5
5. Dataset Preprocessing	7
6. Dataset Splitting	8
7. Model Training & Testing	8
8. Model Selection and Comparison	8
9. Conclusion	11

1. Introduction

This project uses machine learning models to predict the price category of flats based on various features such as location, size, number of bedrooms, and amenities. In a growing urban environment, having a reliable model to estimate property prices can be beneficial for both buyers and sellers. The motivation behind this project is to use machine learning techniques to build a predictive model that can assist in real estate pricing decisions, improving transparency and accuracy in the market. The core problem is accurately classifying flats into price ranges using models like K-Nearest Neighbors (KNN), Logistic Regression, Naive Bayes and Multi-Layer Perceptron (MLP).

2. Dataset Description

- **Dataset:** flat_price_dataset.csv
- **Number of Features:** 11 features (both categorical and numerical)

0	Location	1060	non-null	object
1	Size_sqft	1094	non-null	float64
2	Num_Bedrooms	1064	non-null	float64
3	Num_Bathrooms	1057	non-null	float64
4	Has_Balcony	1079	non-null	object
5	Floor_Number	1068	non-null	float64
6	Building_Age_Years	1073	non-null	float64
7	Parking_Available	1099	non-null	object
8	Nearby_Schools	1076	non-null	object
9	Distance_to_CityCenter_km	1097	non-null	float64
10	Security_Level	1074	non-null	object

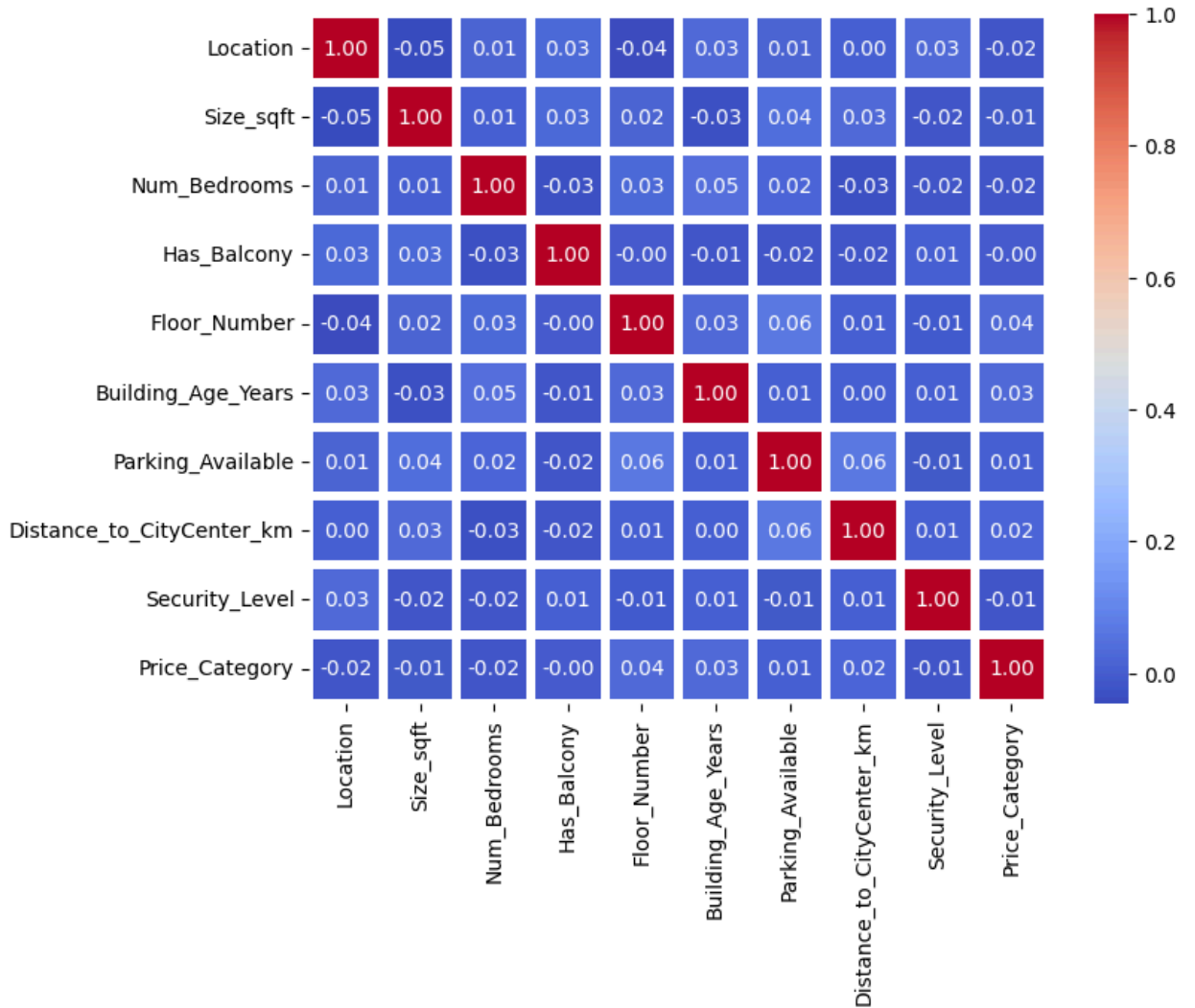
- **Problem Type: Classification**

This is a classification problem as the output feature Price_Category consists of three discrete classes, such as Low, Medium, and High, rather than continuous numerical values.

- **Number of Data Points:** 1200
- **Feature Types:**
 - **Quantitative:** Size_sqft, Num_Bedrooms, Num_Bathrooms, Floor_Number, Building_Age_Years, Distance_to_CityCenter_km

- **Categorical:** Location, Has_Balcony, Parking_Available, Nearby_Schools, Security_Level

- **Correlation Analysis:**



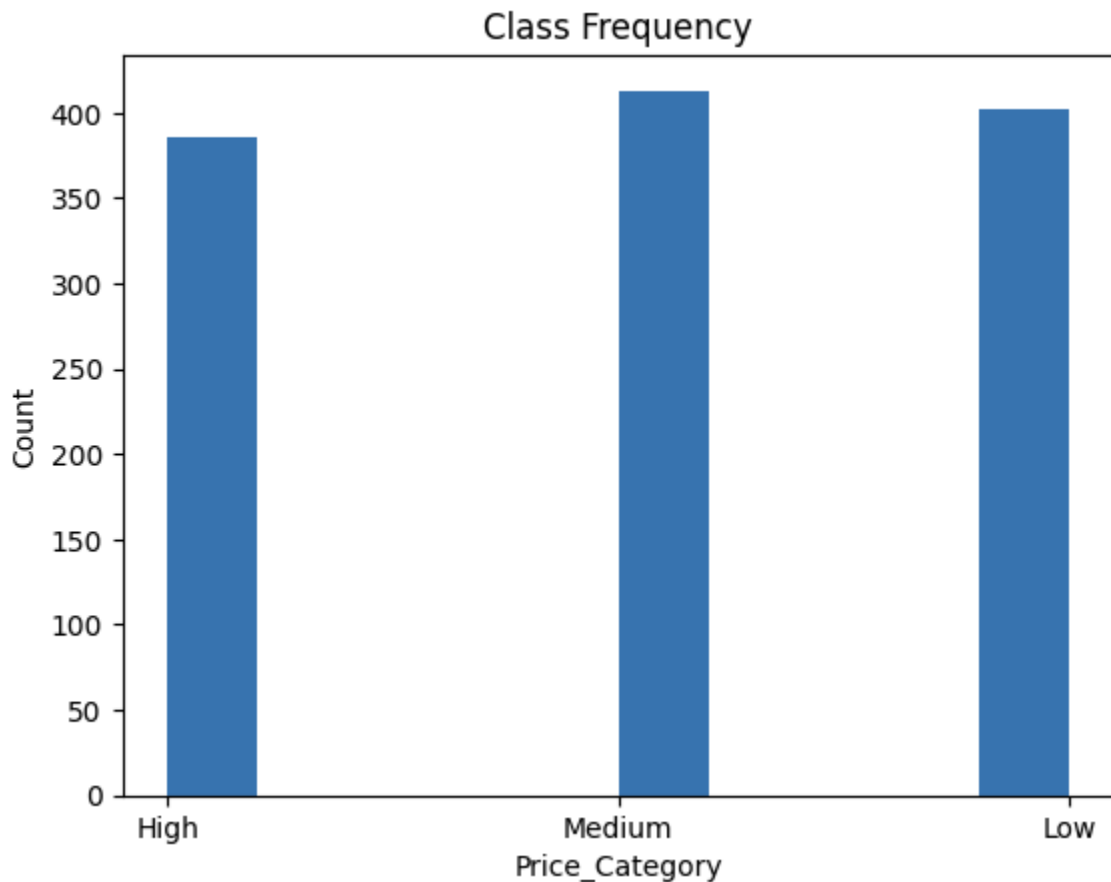
A correlation heatmap was generated using seaborn to examine relationships between features. The heatmap revealed weak correlations between input features as no pair of features exhibits a correlation higher than 0.7 or lower than -0.7, indicating that no severe multicollinearity exists.

While some features showed slight positive or negative relationships, the correlations were too weak to justify removing any of them. As a result, all features were kept for

further analysis and model training, since they might still carry useful information in non-linear ways.

3. Class Distribution and Imbalance

- **Target Feature:** Price_Category
- **Class Balance:** The classes were visualized using a histogram.

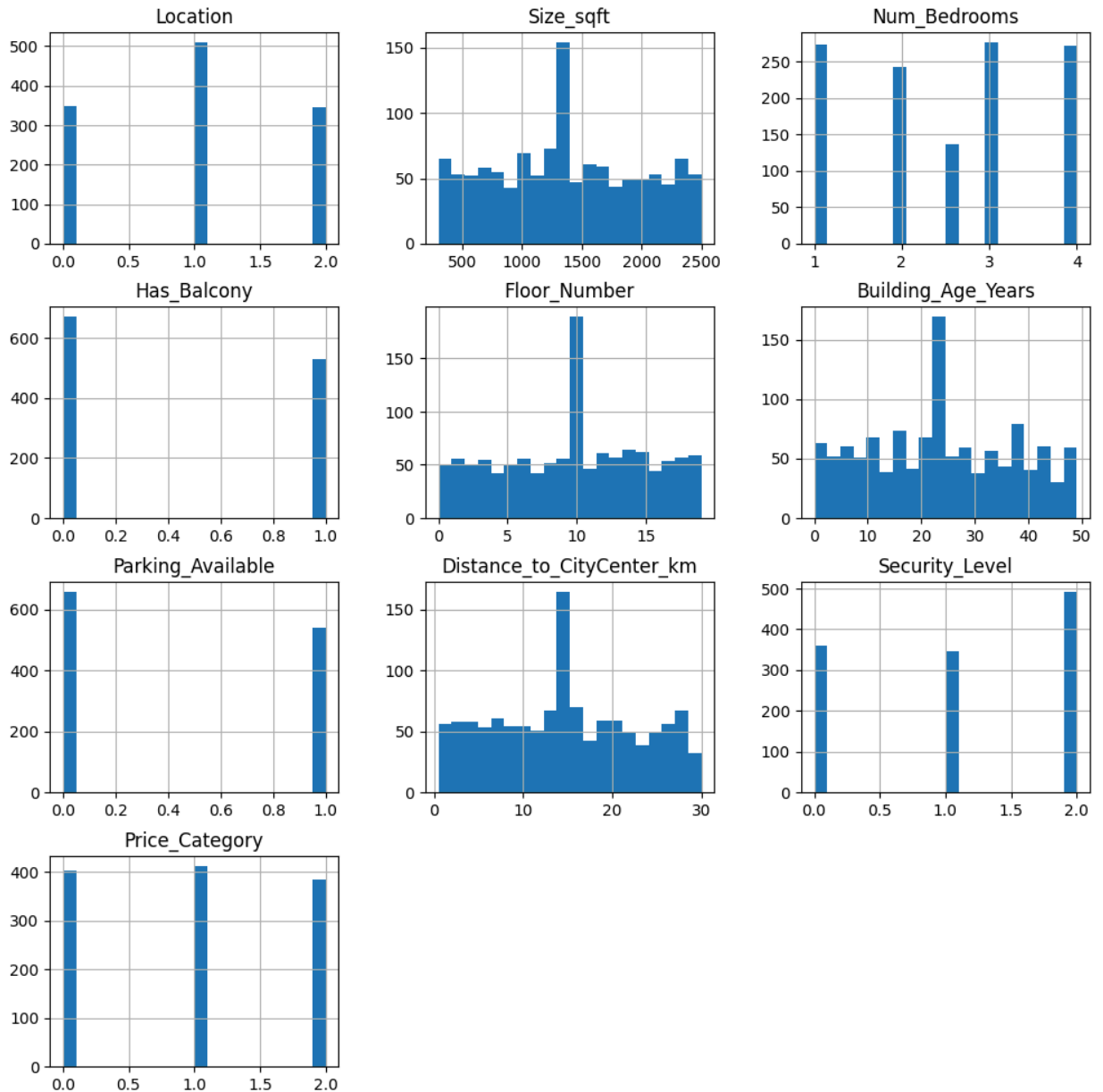


The dataset appears slightly imbalanced, with certain categories having more instances than others. This slight imbalance is unlikely to significantly affect model performance. However, it's still important to keep an eye on the performance metrics like precision, recall, and the confusion matrix.

4. Exploratory Data Analysis (EDA)

- Plotted distributions of individual features.
- Analyzed feature relationships with Price_Category
- Stronger predictive features include Size_sqft and Distance_to_CityCenter_km.

- Checked the skewness of numerical features to understand their distribution better. Skewness indicates whether a feature is symmetrically distributed or biased toward one side. Here, we can see that skewness values are close to zero, indicating that the numerical features are approximately symmetric. As a result, mean imputation is appropriate for handling missing values in these columns.



Skewness of Numerical Features:	
Distance_to_CityCenter_km	0.061188
Building_Age_Years	0.058064
Num_Bathrooms	0.054632
Size_sqft	0.048548
Num_Bedrooms	-0.037031
Floor_Number	-0.074779

5. Dataset Pre-processing

Problem 1: Missing Values

- The dataset contained missing values in both categorical and numerical columns.
- **Solution:**
 - Categorical: Imputed using mode.
 - Numerical: Imputed using the mean since the data was symmetrically distributed.

Problem 2: Categorical Encoding

- Machine learning models cannot interpret categorical text data directly.
- **Solution:** Categorical columns were encoded using Label Encoding . In Label Encoding, each category of a feature is mapped to a unique integer. For example, for Location: 'Countryside' is mapped to 0, 'City Center' to 1, and 'Suburbs' to 2.

Problem 3: Feature Scaling

- Several models used in the project are sensitive to the scale of input features.
- **Solution:** Applied feature scaling using Standardization (Normalization) to numerical features for ensuring consistent scaling. This prevents certain features from dominating due to larger ranges.

Problem 4: Irrelevant Features

- Some features had a large number of missing values and very weak correlation with the target variable.
- **Solution:** Features like Nearby_Schools and Num_Bathrooms contained many missing values and showed very weak correlation with the target (-0.05), this indicates minimal predictive value. Also, the model tests showed a slight improvement in accuracy after dropping them

Missing values per column:	
	0
Location	140
Size_sqft	106
Num_Bedrooms	136
Num_Bathrooms	143
Has_Balcony	121
Floor_Number	133

6. Dataset Splitting

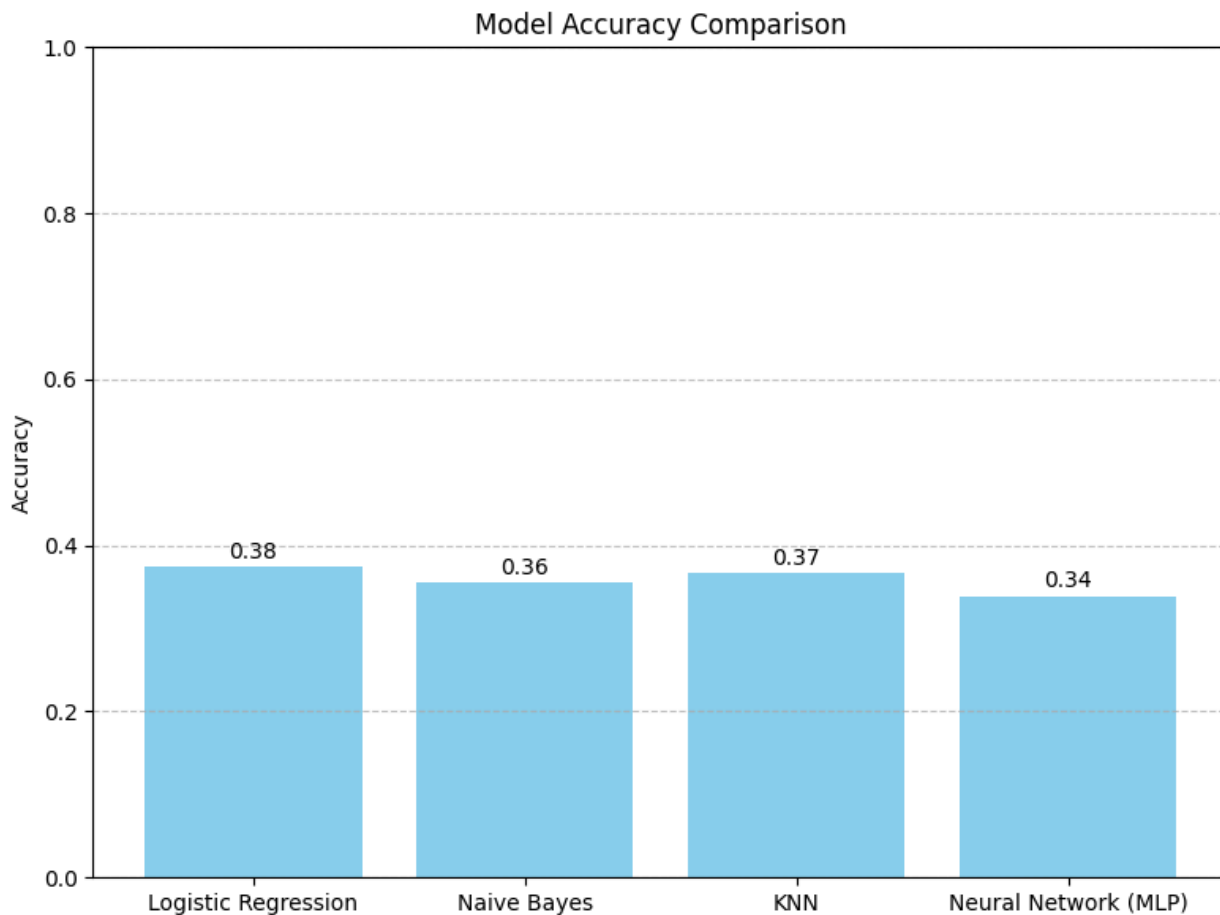
- **Split Method:** Stratified splitting. Even though the class imbalance in our dataset is small, using stratified splitting helps to preserve class proportions between the training and test sets.
- **Training Set:** 70%
- **Testing Set:** 30%

7. Model Training & Testing

- **K-Nearest Neighbors (KNN):** Used to classify based on neighborhood similarity. Performed moderately.
- **Logistic Regression:** Provided interpretable results and decent accuracy.
- **Naive Bayes:** Fast but less accurate due to strong independence assumptions.
- **MLP Techniques:** Handle complex feature interactions effectively.

8. Model Selection/Comparison Analysis

- **Accuracy Bar Chart:** Using matplotlib.pyplot in collab



- **Model Performance Summary:**

Model	Accuracy	Precision	Recall	F1 Score	AUC Score
Logistic Regression	0.3750	0.3897	0.3750	0.3643	0.5080
Naive Bayes	0.3556	0.3586	0.3556	0.3472	0.5097
KNN	0.3667	0.3758	0.3667	0.3607	0.5211
MLP	0.3389	0.3366	0.3389	0.3370	0.5152

```

Evaluation for Logistic Regression
-----
Confusion Matrix:
[[49 57 15]
 [47 62 15]
 [35 56 24]]

Classification Report:
              precision    recall  f1-score   support

     0       0.37         0.40         0.39         121
     1       0.35         0.50         0.41         124
     2       0.44         0.21         0.28         115

```

```

Evaluation for Naive Bayes
-----
Confusion Matrix:
[[44 56 21]
 [45 59 20]
 [42 48 25]]

Classification Report:
              precision    recall  f1-score   support

     0       0.34         0.36         0.35         121
     1       0.36         0.48         0.41         124
     2       0.38         0.22         0.28         115

```

Evaluation for KNN

Confusion Matrix:

```
[[57 44 20]
 [58 47 19]
 [49 38 28]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.35	0.47	0.40	121
1	0.36	0.38	0.37	124
2	0.42	0.24	0.31	115

Evaluation for Neural Network (MLP)

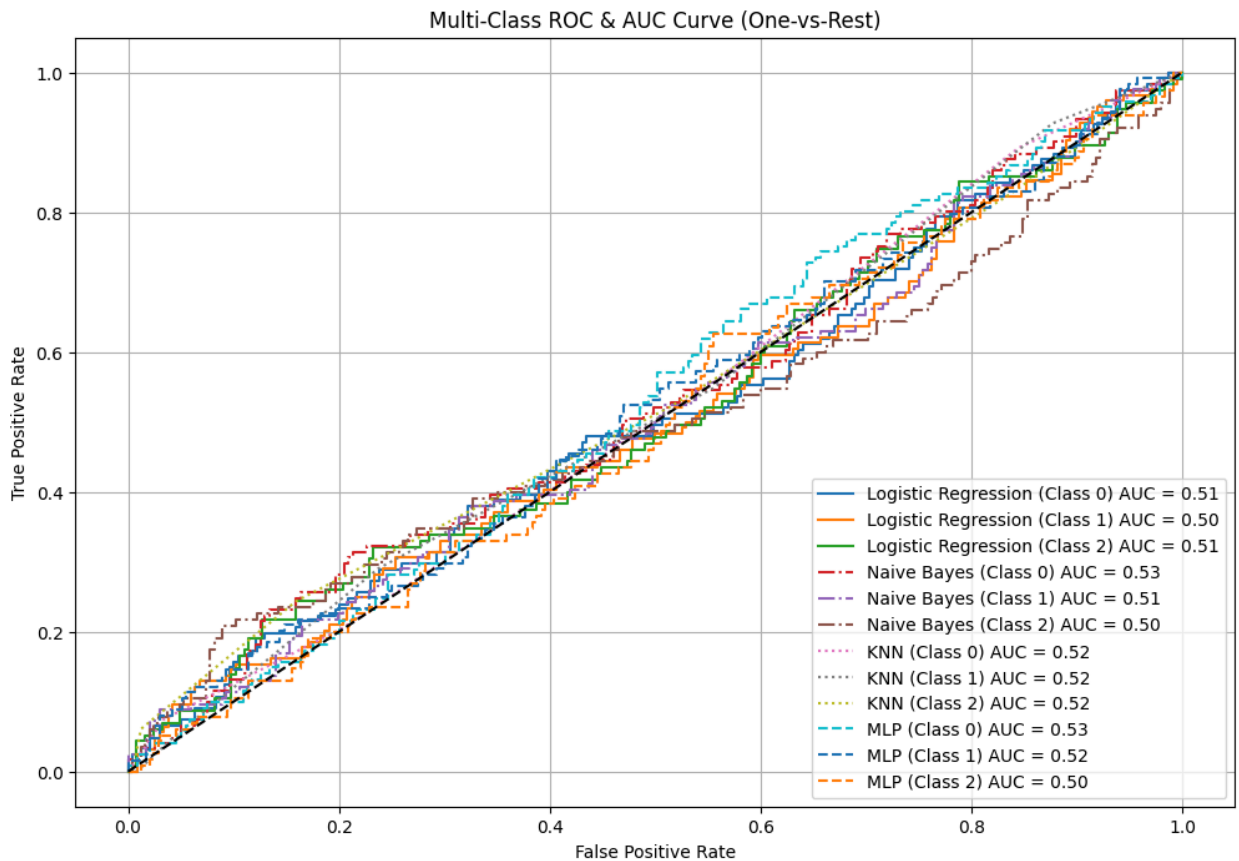
Confusion Matrix:

```
[[43 43 35]
 [41 49 34]
 [44 41 30]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.34	0.36	0.35	121
1	0.37	0.40	0.38	124
2	0.30	0.26	0.28	115

- **ROC Curve & AUC Score:** Evaluated for multiclass



- **Summary:**

- **Logistic Regression** has the highest accuracy, precision, recall, and F1 score, making it the best model overall.
- **KNN** has the highest AUC score.
- **MLP** performed lower overall.

9. Conclusion

This project showed that simpler models like Logistic Regression and KNN can outperform more complex models like MLP on structured datasets. Logistic Regression delivered the best overall performance, while KNN had the strongest AUC score. Some key challenges were handling missing values, imbalanced classes and boosting accuracy through proper preprocessing. Overall, the experience highlighted the importance of EDA, feature engineering and smart model selection when working with real-world datasets.